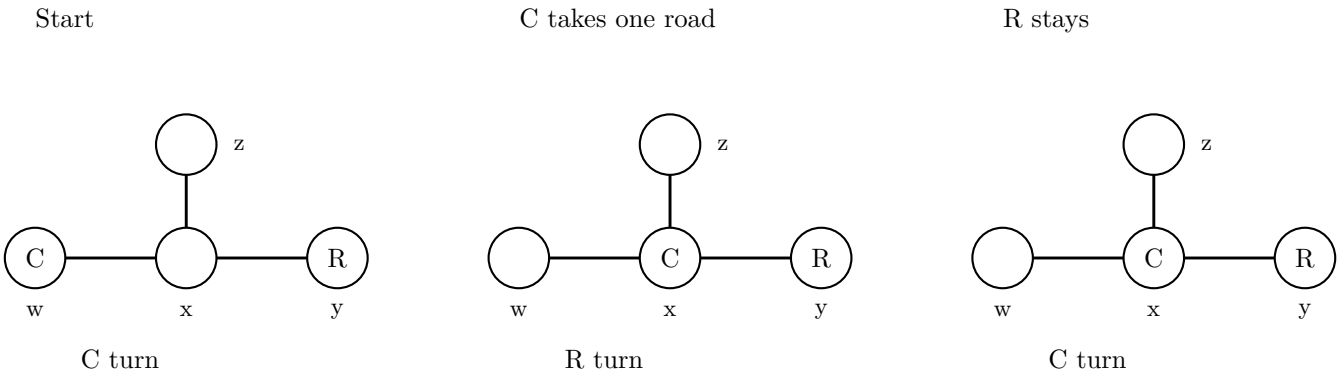
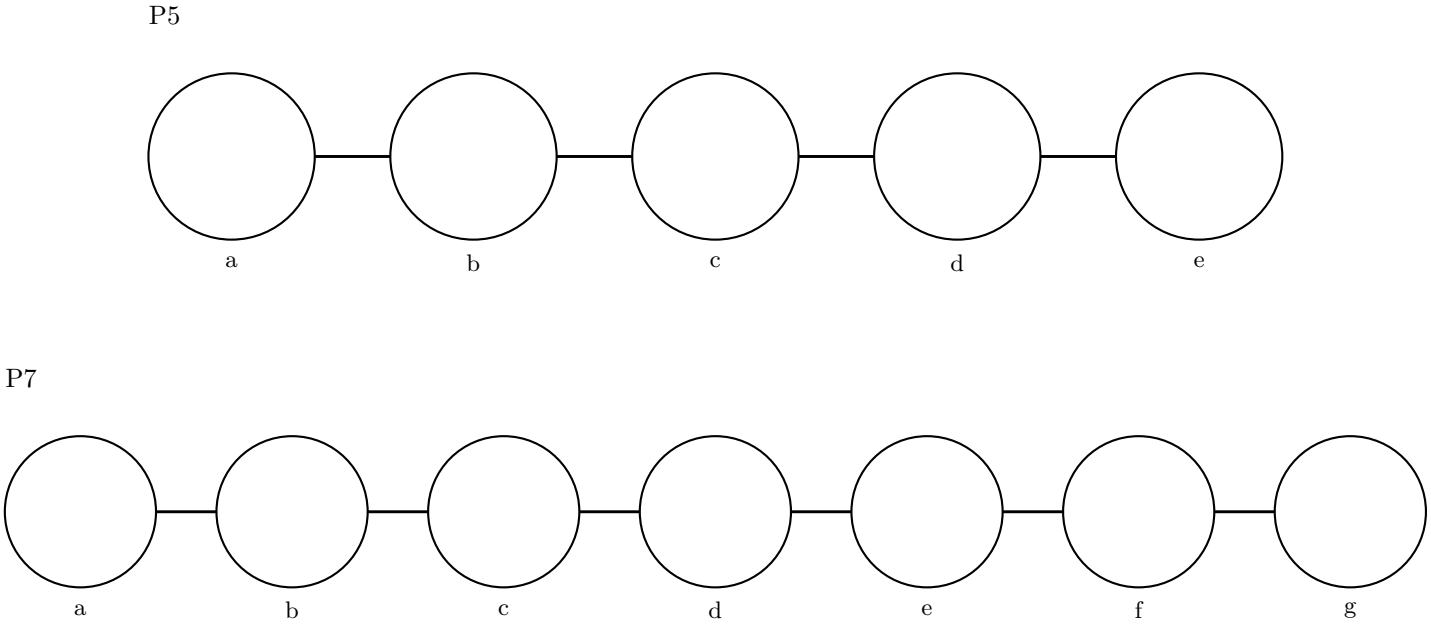
One player controls C; the other controls R. C chooses a dot first, then R chooses. C takes the first turn. On your turn, stay or take one road to the next dot. Flip the turn tile even if you stay. Sharing a dot means immediate capture. Both counters stay visible. Swap jobs between games.

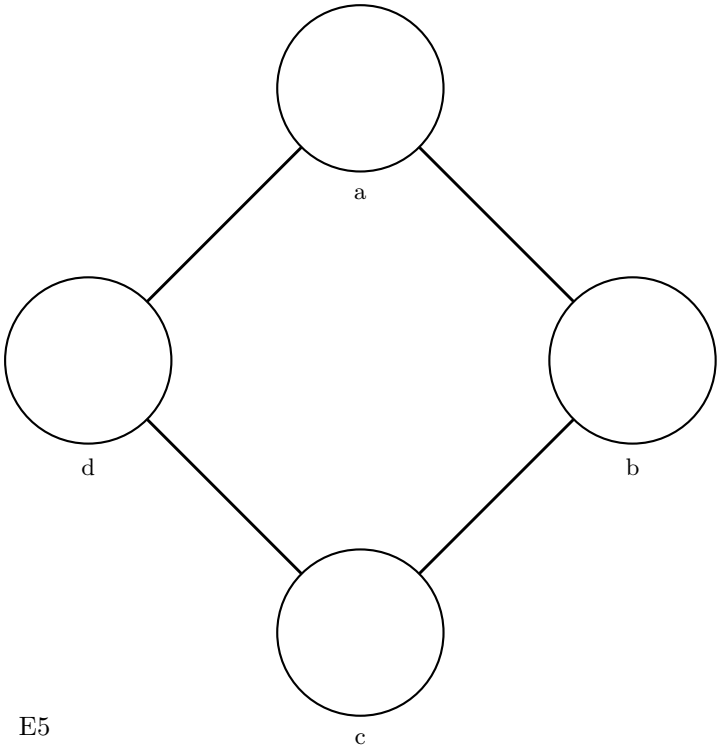


Problem 1: Play on both paths. Choose a starting dot for C that guarantees capture in the fewest C turns, even when R chooses its hardest starting dot and moves.

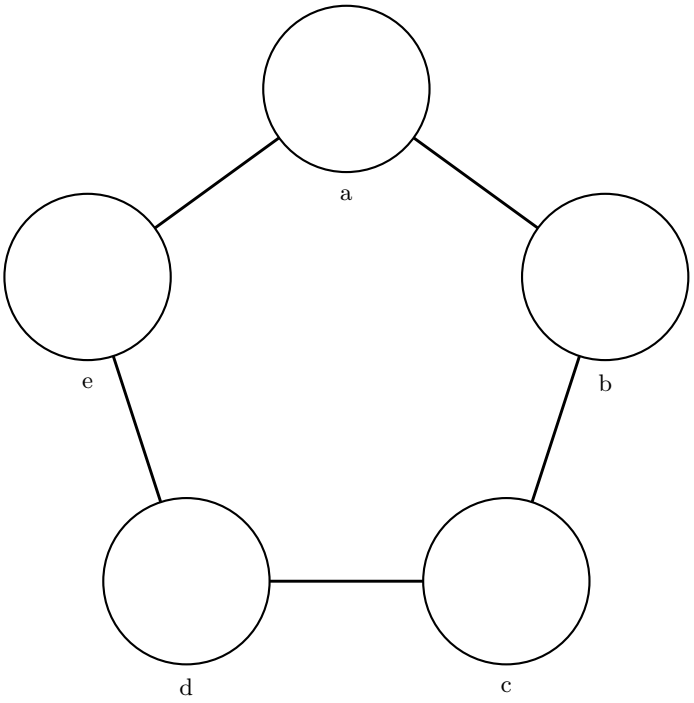


Problem 2: Add as few roads as possible to each ring so C can always catch R. A road joins two existing dots. A crossing is not a new dot. Show a winning cop plan for each changed map.

E4

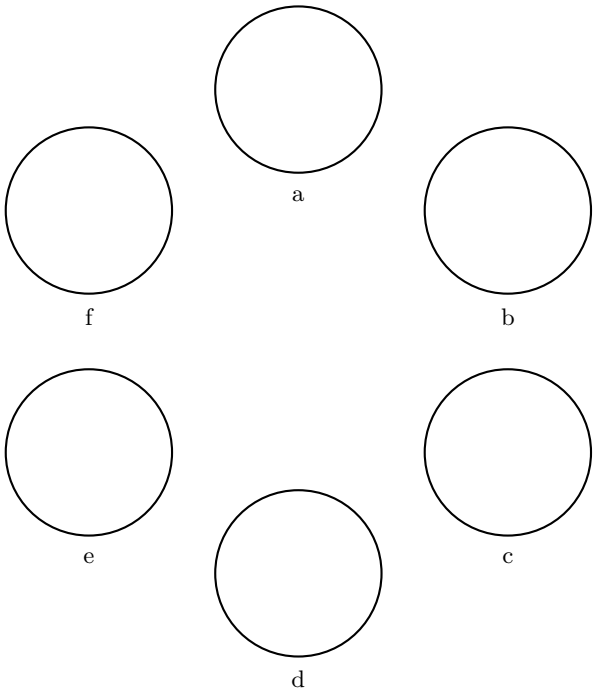


E5

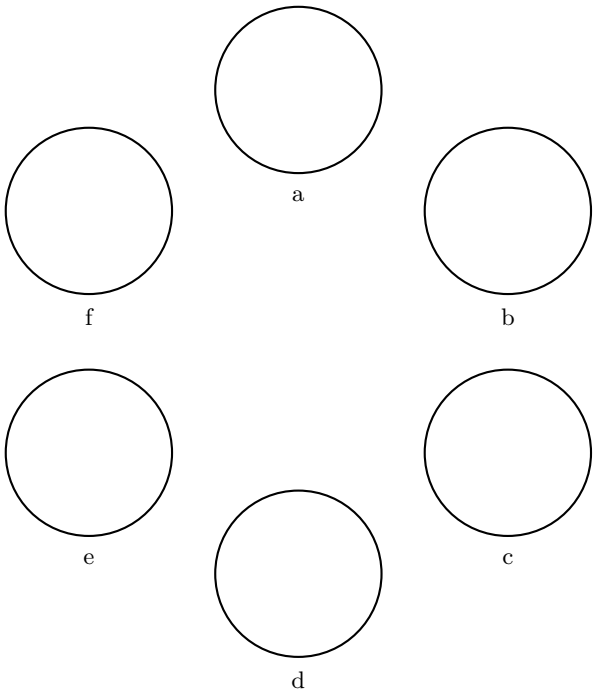


Problem 3: Draw two connected maps using all six dots and at least six roads on each. Make one where C can always catch R and one where R can escape forever. Challenge your partner on both. Roads join dots; a crossing is not a new dot. Join any pair at most once.

D1: C can always catch R

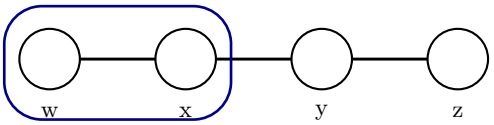


D2: R can escape forever



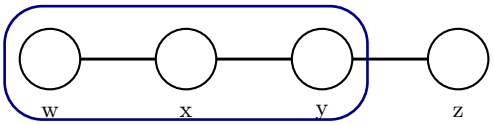
You may remove a dot v when a neighboring dot u can reach every place v can reach by staying or taking one road. This is a fold, written v → u. Remove v and all its roads. Test the next fold on the remaining map.

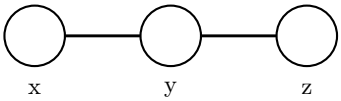
Original: w can reach w, x.



w → x: remove w and road w–x.

Compare: x can reach w, x, y.



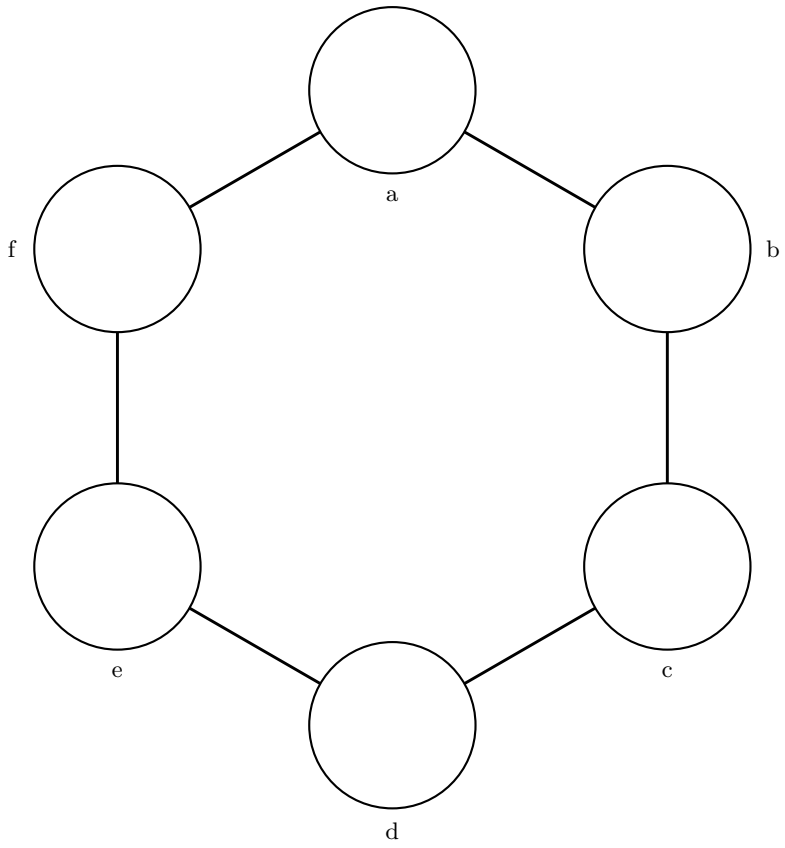


Problem 4: Use boards F1 and F2 in the materials packet. Keep the original uncovered; on the working copy, cover each removed dot and its roads. Can each map fold down to one dot? Find a sequence of legal folds, or a remaining map with no legal fold. Who can always win the chasing game on F1, and on F2?

Use C1 and C2. Put down both cops before R chooses. On a cop turn, each cop may stay or take one road. Then R gets one turn. Flip the tile after the whole cop turn and after R's turn. Capture is immediate. Cops may share a dot.

Problem 5: Start both cops at a. Use them to catch R on this ring. Find a plan that works however R starts and moves.

Q6



Problem 6: Find a two-cop plan for this ring. Will your plan work on every ring, however many dots it has? Explain why.

Q8

